

Computer Science & IT

Discrete Mathematics

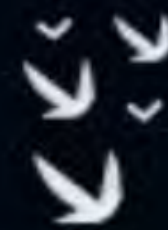
Mathematical Logic

Lecture No. 05



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Recap of Previous Lecture



Topic

Proof by contradiction

Topic

Conditional proof rule

Topic

Some important propositional statements

Topic

Predicate

Topic

Quantifiers

Topics to be Covered



Topic

Quantifiers



Topic

Scope of the quantifier



Topic

Negation of a statement formula



Topic

Sentences with multiple quantifiers



Topic

Relationship diagram



P : is a politician

S : is a sportsman

$$\textcircled{1} \quad \forall x \{ P(x) \rightarrow S(x) \}$$

We are checking whether for all 'x' in the domain $P(x) \rightarrow S(x)$ is true or not
→ Truth value of the predicate formula will be the truth value returned by quantifier $\forall$ {for all}

for all x, if P is true then S is true
for all x, whenever P is true S must be true
 $\forall x \{ P(x) \rightarrow S(x) \}$ will return true if and only if $P(x) \rightarrow S(x)$ is true for all elements in domain
→ If there exists at least one person who is a politician but not a sportsman, then $P(x) \rightarrow S(x)$ will be false, and hence $\forall$ will return false

$$\textcircled{2} \quad \underbrace{\forall x \{ P(x) \}}_{\text{L.H.S.}} \rightarrow \underbrace{\forall x \{ S(x) \}}_{\text{R.H.S.}}$$

Truth value of the predicate formula is the truth value returned by implication

→ In this case implication is the main part of the predicate formula, so if LHS is true then R.H.S. must be true for implication to return true
→ if L.H.S. then R.H.S.
i.e. if all are politician then all must be sportsman
→ if there exist at least one person who is not a politician then $\forall x \{ P(x) \}$ will return false, i.e. L.H.S. of implication will become false, hence implication will return true

H.W. Which of the following Predicate formula is/are valid

$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$

$U = \{1, 2, 3\}$

$P(1) = F$ $P(2) = T$ $P(3) = T$	$\therefore \forall x \{P(x)\} = \text{False}$ i.e. L.H.S. is false $\therefore$ implication return true	$S(1) = T$ $S(2) = F$ $S(3) = T$	$P(1) \rightarrow S(1) = T$ $P(2) \rightarrow S(2) = F$ $P(3) \rightarrow S(3) = T$	$\therefore \forall x \{P(x) \rightarrow S(x)\}$ will return false
L.H.S.			R.H.S.	
True			False	

$\text{True} \rightarrow \text{False} = \text{False} \{ \text{Hence invalid} \}$

$F_2: [\forall x \{P(x) \rightarrow S(x)\}] \rightarrow [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}]$

H.W. Which of the following Predicate formula is/are valid

$$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$$

it is invalid

$$F_2: [\forall x \{P(x) \rightarrow S(x)\}] \rightarrow [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}]$$

is R.H.S. will also be true

We are saying.
For all x , Whenever P is true
 S must be true
i.e., if at all ' P ' happens to be true
for all ' x ' then S will also be
true for all x
i.e. if $\forall x \{P(x)\}$ is true then $\forall x \{S(x)\}$ will
also be true
is valid.

if P happens to be false
for at least one case, then
 $\forall x \{P(x)\}$ will return false
And hence $\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}$
will always be true
i.e. R.H.S. of main implication will be true
And hence the main implication
will return true

so F_2
is valid



Topic : Existential Quantifier ($\exists$)

$\exists$: for some { for at least one }

$\exists x$: for some 'x' { for at least one 'x' }
in domain { in domain }

P : is a politician

$\exists x \{ P(x) \}$: Some are Politician
(or)

At least one person in the set
is a politician

it may be true, it may be false

$\exists$ will return true if and only if at least one
person in the domain is a politician

$\exists$ will return true if and only if Predicate under the scope is true for at least one element in the domain

$\therefore$ if domain is empty then

$\exists$ will always return false

$\exists x \{ P(x) \}$

quantifier

scope of quantifier

$\exists x \{ P(x) \rightarrow S(x) \}$

scope of quantifier



Topic : Scope of the Quantifier

- The part of the logical expression to which a quantifier can be applied is called the scope of the quantifier.
- A variable whose occurrence is bounded by a quantifier is called a bounded variable. Variables not bounded by any quantifiers are called free variables.

* Generally we represent the scope of the quantifier using brackets or comma otherwise, The Scope of a quantifier is the shortest full sentence/predicate formula which follows it. Everything inside this shortest full sentence is said to be in the scope of the quantifier.



Topic : Scope of the Quantifier

- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $\exists y(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$
- $(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$



Topic : Scope of the Quantifier

- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$

'y' is a bounded Variable

'z' is a bounded Variable

free Variable

bounded Variable



Topic : Scope of the Quantifier



A variable can not be quantified using multiple quantifiers

Even if it seems like that the variable is bounded by multiple quantifiers, then it is actually bounded by inner most quantifier

$$\exists y (\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$$

all free occurrences of y under the scope will be quantified using " $\exists y$ "

under the scope of $\forall y$

bounded variable

bounded variable

under the scope of $\exists y$
(bounded variable)

Same formula can be represented using the expression $\exists x \{ \forall y H(y) \rightarrow \exists z (z, x) \} \rightarrow \exists z (G(z))$



Topic : Scope of the Quantifier

- $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$
The diagram illustrates the scope of quantifiers in the logical expression $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$.
 - Brackets and arrows labeled "bounded" indicate the scope of the quantifiers $\forall y$, $\exists z$, and $\exists z$.
 - A pink arrow labeled "free variable" points to the variable y in the predicate $W(z, y)$, indicating it is free.



Topic : Scope of the Quantifier

$$(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$$

for all will
quantify
Variable 'z'

free
Variable

bounded

free
Variable

bounded



Topic : Scope of the Quantifier

• $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$

Diagram illustrating the scope of quantifiers in the formula $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$:

- The scope of $\forall y$ is $H(y)$ (indicated by a pink bracket and the word "bounded" in pink).
- The scope of $\exists z$ is $\exists y W(z, y)$ (indicated by a blue bracket and the word "bounded" in blue).
- The scope of $\exists y$ is $W(z, y)$ (indicated by a green bracket and the word "bounded" in green).
- The scope of the outer $\exists z$ is $G(z)$ (indicated by a green bracket and the word "bounded" in green).



Topic : Scope of the Quantifier

• $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$

Diagram illustrating the scope of quantifiers in the formula $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$:

- $\forall y$ is bounded (scope: $H(y)$).
- $\exists z$ is bounded (scope: $W(z, y)$).
- $\exists x$ is free (scope: $G(z)$).

Handwritten annotations in the image indicate the scope of each quantifier with brackets and arrows:

- A blue bracket above $\forall y$ and an arrow pointing to $H(y)$ are labeled "bounded".
- A blue bracket above $\exists z$ and an arrow pointing to $W(z, y)$ are labeled "bounded".
- A pink bracket above $\exists x$ and an arrow pointing to $G(z)$ are labeled "free".



Topic : Note

- Let, $P(x)$: x is true.

$\sim P(x)$: x is not true
i.e. x is false



	Form	Meaning
1	$\forall x P(x)$	All are true $\equiv$ Not at least one is false
2	$\exists x P(x)$	At least one true $\equiv$ Not all are false
3	$\sim (\forall x P(x))$	Not all true $\equiv$ At least one is false
4	$\sim (\exists x P(x))$	Not at least one true $\equiv$ All are false
5	$\forall x (\sim P(x))$	All are false $\equiv$ Not at least one is true
6	$\exists x (\sim P(x))$	At least one false $\equiv$ Not all are true
7	$\sim (\forall x \sim P(x))$	Not all are false $\equiv$ At least one is true
8	$\sim (\exists x \sim P(x))$	Not At least one false $\equiv$ All are true



Topic : Equivalences

1. $\forall x P(x) \equiv \sim \exists x (\sim P(x))$

$\exists x$ replaced by $\forall x$

Scope of quantifier is negated
i.e. $\sim(\sim P(x)) \equiv P(x)$

2. $\exists x P(x) \equiv \sim \forall x (\sim P(x))$

$\exists x \{P(x)\}$

3. $\sim \forall x P(x) \equiv \exists x (\sim P(x))$

$\exists x \{ \sim P(x) \}$

4. $\sim \exists x P(x) \equiv \forall x (\sim P(x))$

$\forall x \{ \sim P(x) \}$



Topic : Negation of a statement formula

When negation sign is propagated inside the Predicate formula,

then Replace $\forall$ by $\exists$

and
replace $\exists$ by $\forall$

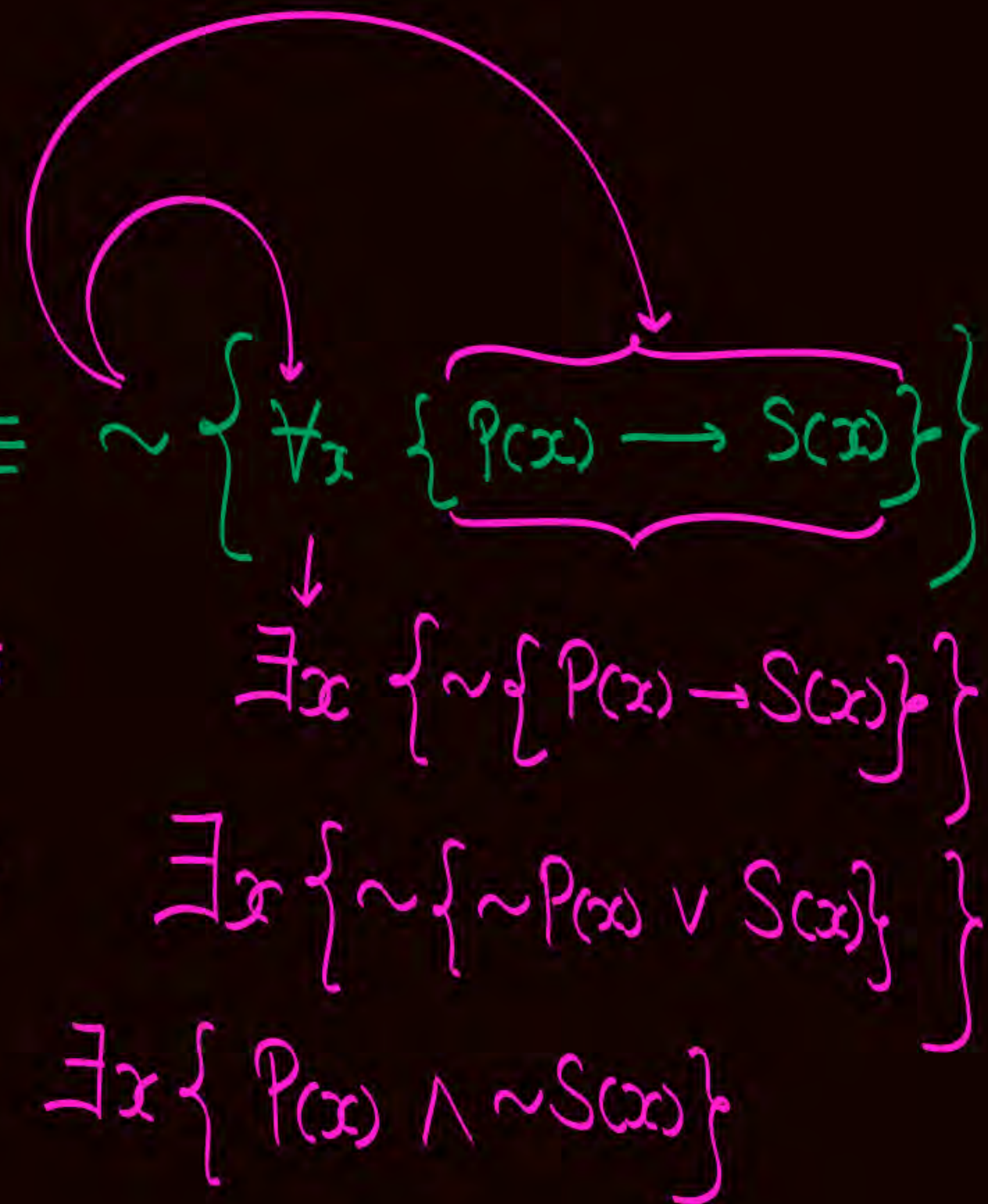
and

negate the scope of the corresponding quantifier

Q. Negate the predicate formula
 $\forall x \{ P(x) \rightarrow S(x) \}$

Solution

Negation of $\forall x \{ P(x) \rightarrow S(x) \}$

$$\begin{aligned} &\equiv \sim \left\{ \forall x \{ P(x) \rightarrow S(x) \} \right\} \\ &\equiv \exists x \{ \sim \{ P(x) \rightarrow S(x) \} \} \\ &\equiv \exists x \{ \sim \{ \sim P(x) \vee S(x) \} \} \\ &\equiv \exists x \{ P(x) \wedge \sim S(x) \} \end{aligned}$$


Q: The predicate formula $\forall x \{ B(x) \wedge I(x) \}$ is equivalent to which of the following predicate formula

- (a) $\exists x \{ B(x) \rightarrow \sim I(x) \}$
- ☒ (b) $\sim \exists x \{ B(x) \rightarrow \sim I(x) \}$
- (c) $\sim \exists x \{ \sim B(x) \rightarrow I(x) \}$
- (d) $\exists x \{ \sim B(x) \rightarrow I(x) \}$

$$A \equiv \sim(\sim A)$$

$$\begin{aligned} \forall x \{ B(x) \wedge I(x) \} &\equiv \sim(\sim \forall x \{ B(x) \wedge I(x) \}) \\ &\equiv \sim \{ \exists x (\sim B(x) \vee \sim I(x)) \} \\ &\quad \boxed{P \rightarrow Q \equiv \sim P \vee Q} \\ &\equiv \sim \{ \exists x (B(x) \rightarrow \sim I(x)) \} \\ &\quad ||| \\ &\equiv \sim \{ \exists x (I(x) \rightarrow \sim B(x)) \} \end{aligned}$$



2 mins Summary



Topic

Quantifiers

Topic

Scope of the quantifier

Topic

Negation of a statement formula

Topic

Sentences with multiple quantifiers

Topic

Relationship diagram

THANK - YOU